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ALCF Inference Service

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Argonne Leadership Computing
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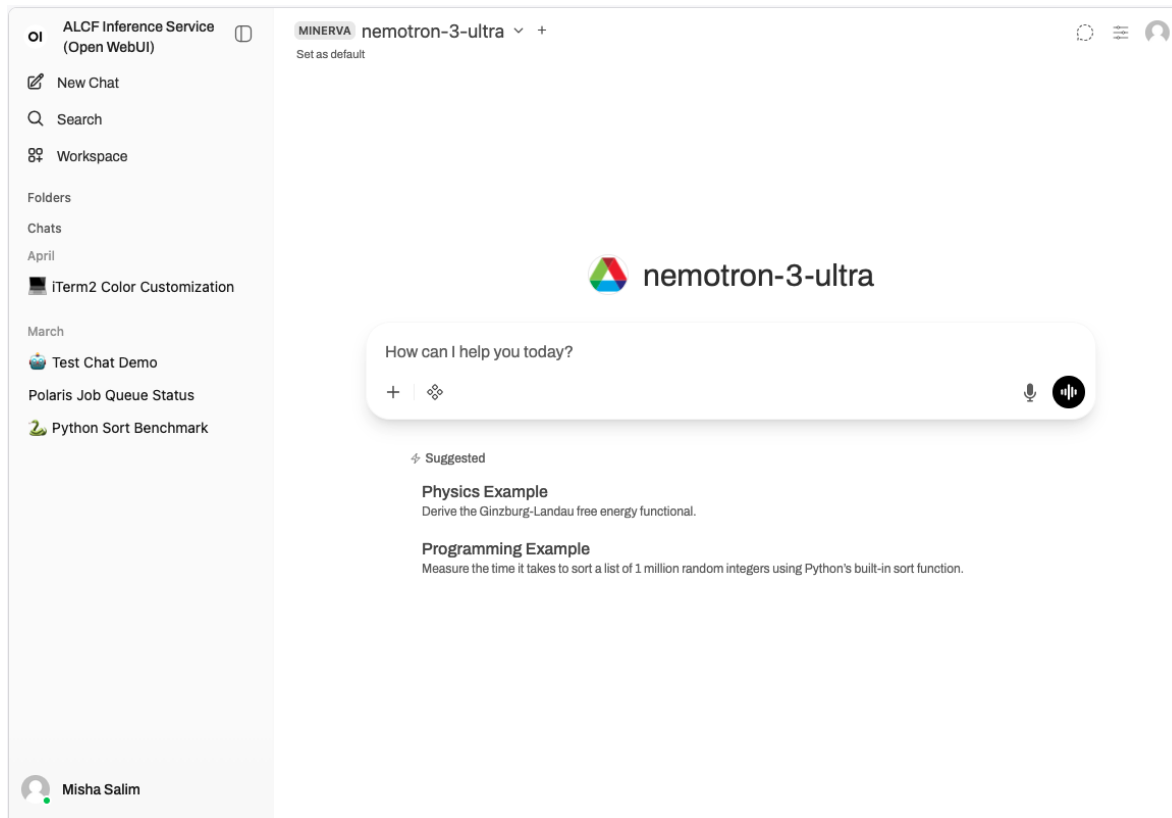
Inference Service

- Always-on, internet-accessible open weight LLMs and scientific models hosted @ALCF
- OpenAI- and Anthropic-compatible APIs provide a foundation for agentic workflows and enable researchers to apply AI to their data, simulations, and real-time experiments.
- Shared service means model lifetime is not tied to your own workloads: just point at <https://inference-api.alcf.anl.gov/> and go
- Log in via Globus with ALCF user account or any other DOE lab credentials
 - *For today's session, use ALCF login for unified access to Inference + other facility services*

Web UI and APIs

<https://inference.alcf.anl.gov>

Chat UI



<https://inference-api.alcf.anl.gov>

OpenAI and Anthropic-compatible APIs

```
#!/bin/bash

# Get your access token
access_token=$(uvx alcf-ai auth get-access-token)

curl -X POST "https://inference-api.alcf.anl.gov/resource_server/metis" \
  -H "Authorization: Bearer ${access_token}" \
  -H "Content-Type: application/json" \
  -d '{
    "model": "gpt-oss-120b",
    "messages": [{"role": "user", "content": "Explain quantum computing"}]
  }'
```

Model Catalog

- ALCF is continually adding newly-released models
- <https://docs.alcf.anl.gov/services/inference-endpoints>
- If you have an LLM or scientific model you'd like hosted at ALCF, please contact support@alcf.anl.gov

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About Computing for Science Allocation Programs and Awards Community and Outreach Genesis Mission

ALCF User Guides

- Running Jobs with PBS
 - Example Job Scripts
 - Common PBS Issues
 - Machine Reservations
- Data Management
 - File System & Storage >
 - Transfer & Sharing >
- Services
 - Inference Endpoints
 - IRI API
 - JupyterHub
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 - Globus Compute
- Account & Project Management
 - MyALCF
 - Accounts and Access >
 - Project Management >
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Available Models ↑ Back to top

Models are organized by cluster and marked with the following capabilities:

- **B** - Batch Processing Enabled
- **T** - Tool Calling Enabled
- **R** - Reasoning Enabled
- **H** - Always Hot Model

Sophia Cluster

- Chat Language Models (vLLM)
- Meta Llama Family**
 - meta-llama/Meta-Llama-3.1-8B-Instruct^{BTH}
 - meta-llama/Meta-Llama-3.1-70B-Instruct^{BTH}
 - meta-llama/Meta-Llama-3.1-405B-Instruct^{BT}
 - meta-llama/Llama-3.3-70B-Instruct^{BT}
 - meta-llama/Llama-4-Scout-17B-16E-Instruct^{BT}
 - meta-llama/Llama-4-Maverick-17B-128E-Instruct^T

A subset of models are always live

Model Catalog

General-Purpose LLMs

- **Llama (Meta)** — 3.1-8B / 70B / 405B-Instruct, 3.3-70B-Instruct, Llama-4 Scout-17B-16E & Maverick-17B-128E
- **Gemma (Google)** — gemma-3-27b-it; gemma-4 26B-A4B / 31B / E4B-it
- **Mistral** — Large-2407, Large-3-675B-2512, Mixtral-8x22B-v0.1, Devstral-2-123B-2512
- **Nemotron-3 (NVIDIA)** — Super-120B, Ultra
- **gpt-oss (OpenAI)** — 20B, 120B
- **Trinity-Large-Thinking (Arcee)** — W4A16 quantized
- **Inklings (Thinking Machines)**

Embedding Models

- embeddinggemma-300m (Google)
- SFR-Embedding-Mistral (Salesforce)
- Mistral-7B-Instruct-v0.3-embed

Vision & Multimodal

- Llama-3.2-90B-Vision-Instruct — multimodal VLM
- SAM 3 — promptable image segmentation
- DINOv3 — SoTA visual representation

Scientific Models

- AuroraGPT (Argonne) — IT-v4, Tulu3-SFT, DPO-UFB, KTO-UFB checkpoints
- AstroSage-70B (AstroMLab) — astronomy
- GenSLM — genome-scale language models
- AmSC SUF Partnership — 7 HEP models, Triton backend

LLM Endpoints

/v1/chat/completions
/v1/responses
/v1/messages
/v1/completions
/v1/embeddings

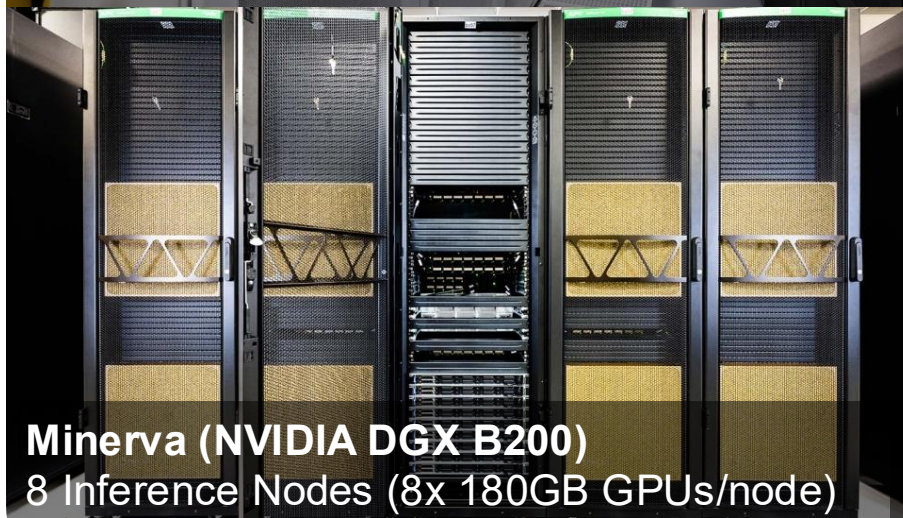
The Hardware



Sophia (NVIDIA DGX A100)
10 Inference Nodes (8x 40GB GPUs/node)



Metis (SambaNova SN40L)
6 Inference Nodes (16 accelerators/node)

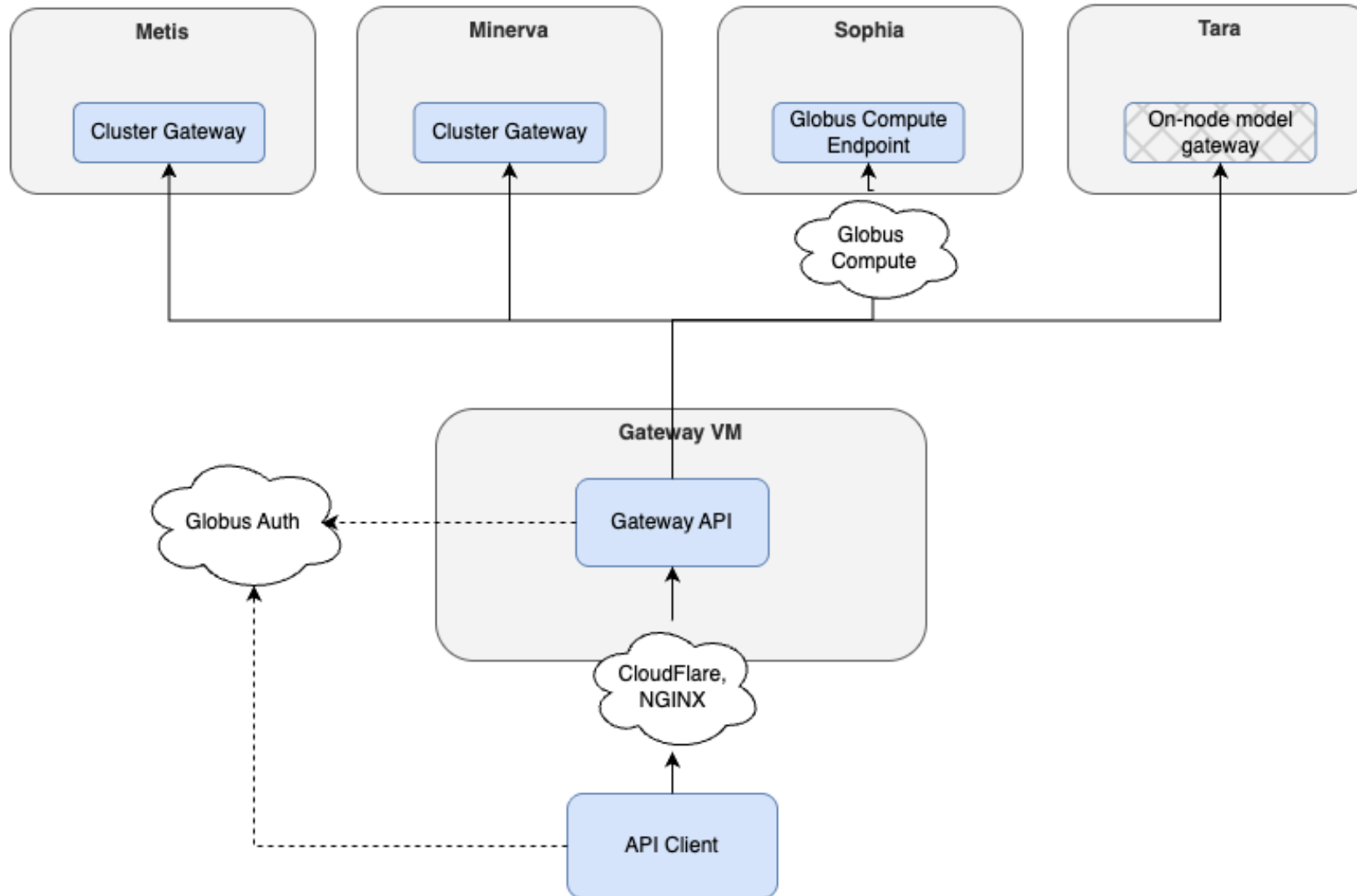


Minerva (NVIDIA DGX B200)
8 Inference Nodes (8x 180GB GPUs/node)



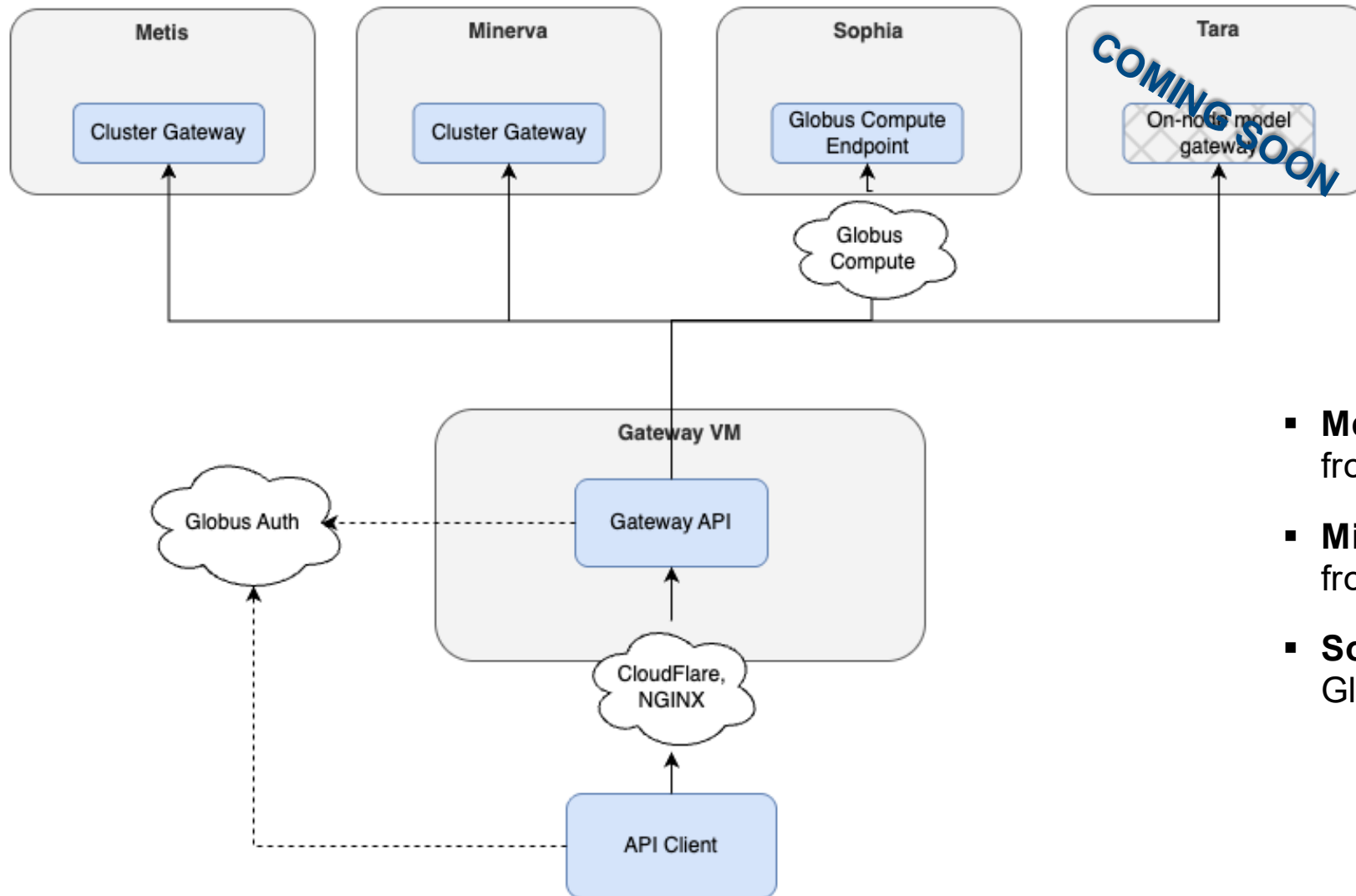
Tara (NVIDIA GH200 / HPE Cray EX254n)
672 Inference Nodes (4x 96GB GPUs/node)

Gateway Overview



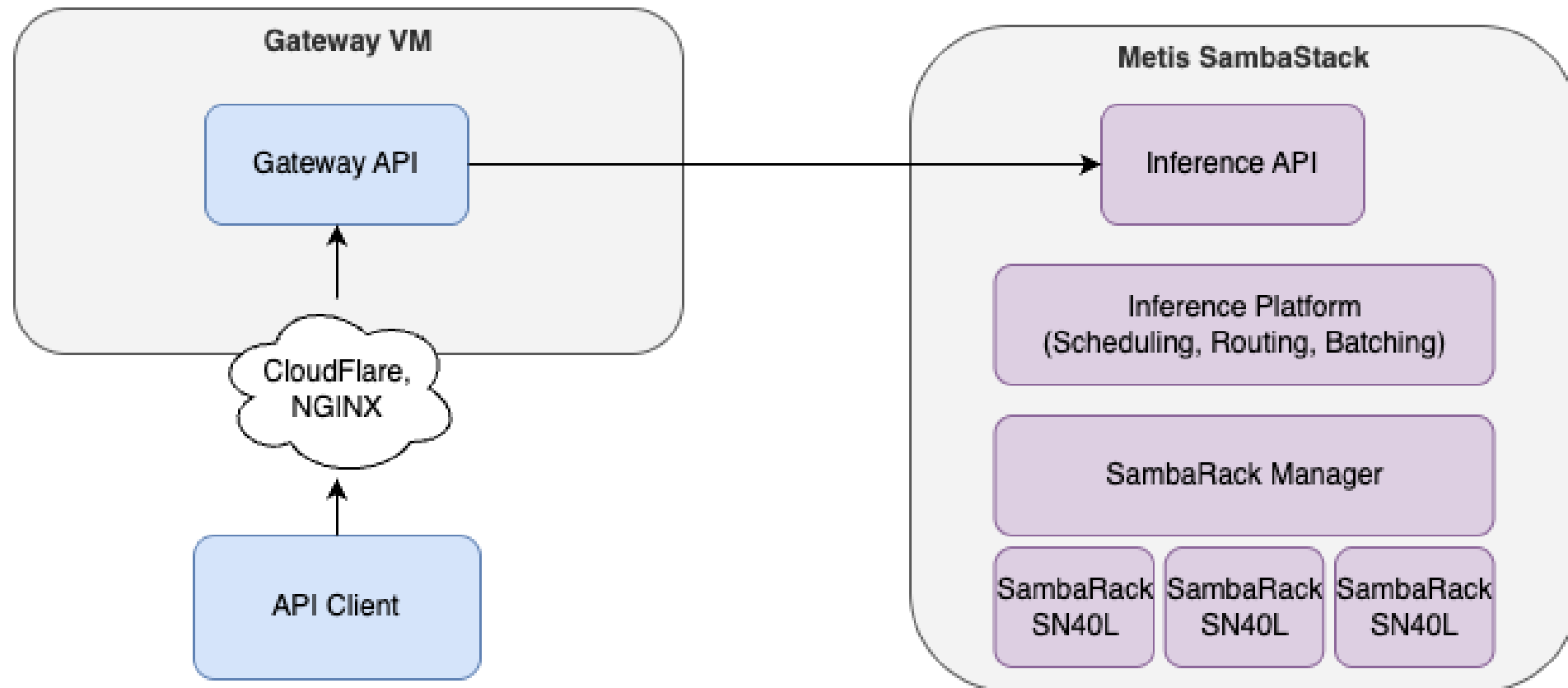
- Globus OAuth2 flows for API Clients and Web UI
- Multi-backend integration
- Live model registry
 - Always-on
 - Autoscaling / scale-to-zero

Backend Integrations

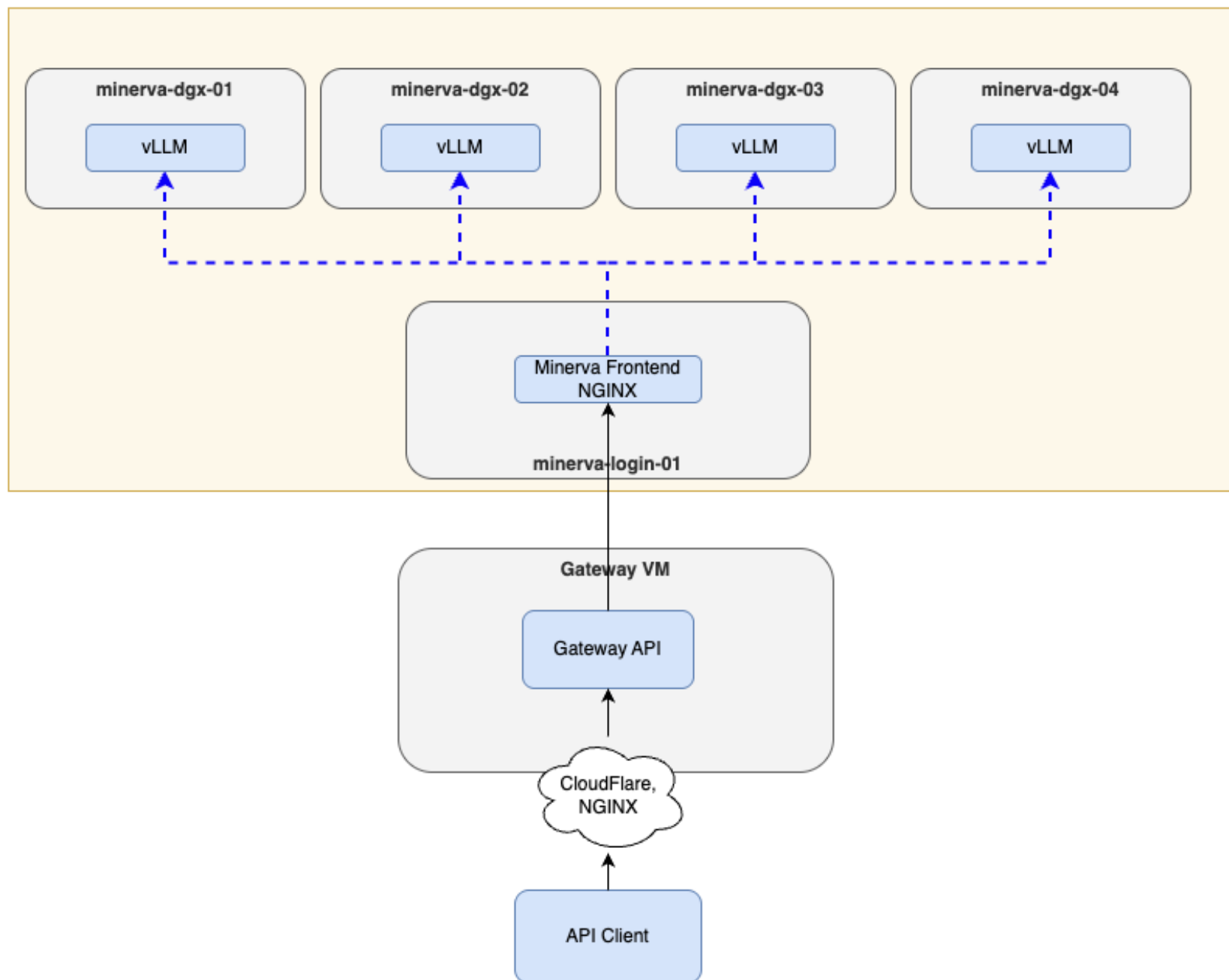


- **Metis:** Direct HTTPS proxy to SambaStack frontend
- **Minerva:** Direct HTTPS proxy to cluster frontend
- **Sophia:** inference requests invoked through Globus Compute

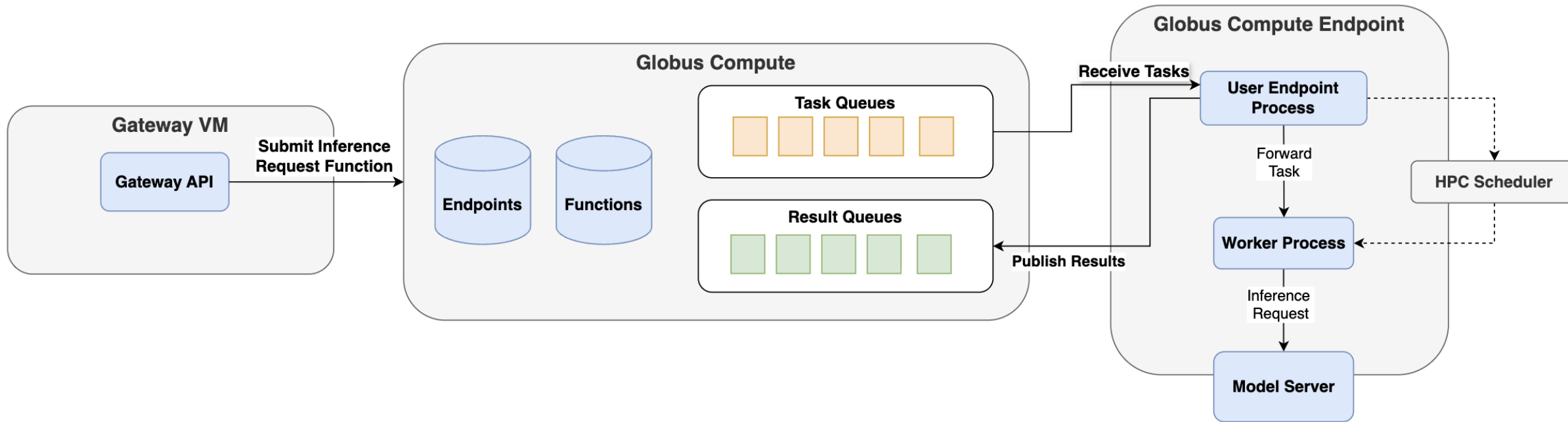
Metis



Minerva



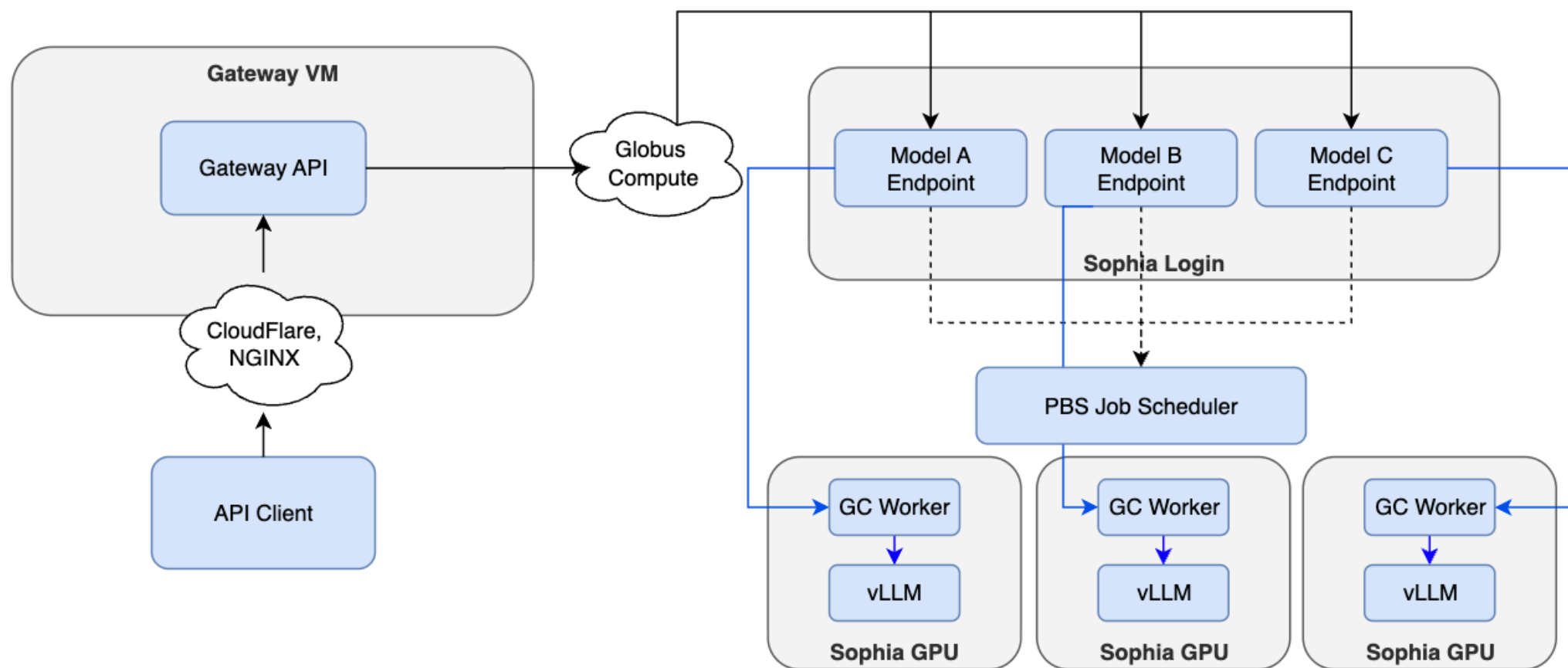
Sophia



- Globus Compute provides a cloud-hosted FaaS platform enabling distributed Python function execution across **endpoints** running anywhere
- The Inference Service adopts this platform to simultaneously provision models, autoscale, and route inference traffic on the Sophia cluster

<https://globus-compute.readthedocs.io/>

Sophia



API Access: Resources

<https://docs.alcf.anl.gov/services/inference-endpoints>

API Access ¶

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For programmatic access, you can use the API endpoints directly.

1. Setup Your Environment

You can run the following setup from anywhere (your local machine, or an ALCF machine).

```
# Create a new Conda environment
conda create -n globus_env python==3.11.9 --y
conda activate globus_env

# Install necessary packages
pip install openai globus_sdk
```

2. Authenticate

To access the endpoints, you need an authentication token.

```
# Download the authentication helper script
wget https://raw.githubusercontent.com/argonne-lcf/inference-endpoints/refs/heads/main/inference_auth_token.py

# Authenticate with your Globus account
python inference_auth_token.py authenticate
```

This will generate and store access and refresh tokens in your home directory. To see how much time before your access token expires, type the following command (`units` can be seconds, minutes, or hours):

```
python inference_auth_token.py get_time_until_token_expiration --units seconds
```

<https://pypi.org/project/alcf-ai/>

alcf-ai 0.7.0

`pip install alcf-ai`

Latest release

Released: Jul 30, 2024

ALCF Inference Gateway SDK

Navigation

- Project description
- Release history
- Download files

Project description

ALCF AI Inference Services SDK

This package provides Python client and CLI tools to facilitate usage of the ALCF AI Inference services.

Command Line Usage

Quick Start

```
# Log in with Globus:
uvx alcf-ai auth login

# Chat with a model
# The default --model is meta-llama/Llama-4-Scout-17B-16E-Instruct
uvx alcf-ai chat "How do I know Pi is irrational? Be concise."
```

Verified details ✓

These details have been [verified by PyPI](#)

Maintainers

 [msalim](#)

Meta

- Author: [Misha Salim](#) ✉

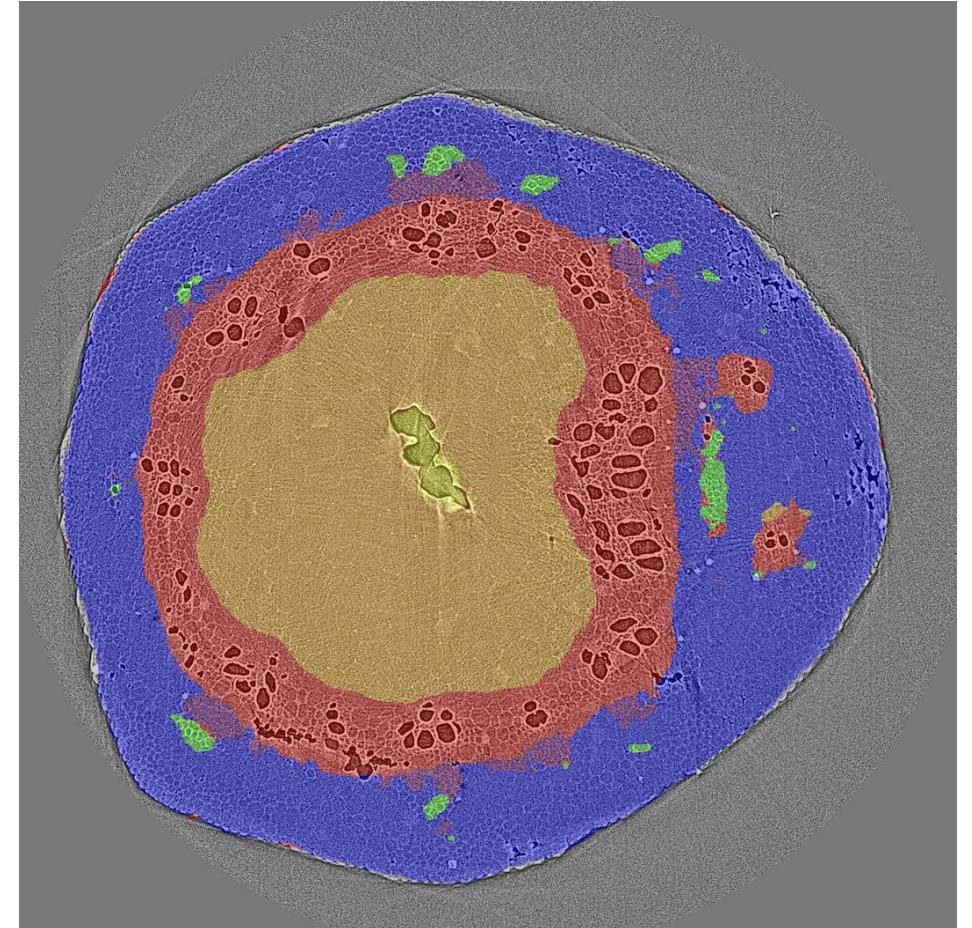
Unverified details

Auth

API Access: Beyond LLMs

- Segmentation Models like **SAM3** and **DINOv3** enable *real-time* AI-assisted segmentation of images generated at DOE light sources
- **alcf-ai** orchestrates bulk image transfers; staging data to/from Inference Service

```
$ uvx alcf-ai dinov3 submit \  
> --origin-collection-id $SOURCE_COLLECTION \  
> --save-overlay \  
> /datascience/msalim/sam3_datasets/APS/SYNAPS-I/example/raw
```



Hands-on examples

- https://github.com/argonne-lcf/Service_Enabled_Science